

BLOG

Photovoltaic Cell Efficiency | PERC vs. TOPCon vs. HJT Technologies

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PERC efficiency 22-24%, low cost, compatible with old production lines, mainstream first choice;
TOPCon relies on tunnel oxide layer to boost efficiency to 24-26%, optimal for transitional upgrade;
HJT low-temperature process + high bifaciality, efficiency 25-27% but expensive, used in high-end scenarios.

PERC

PERC (Passivated Emitter and Rear Cell) is a photovoltaic technology proposed in 1989 by the University of New South Wales, Australia. Through a rear-side aluminum oxide passivation layer + local contact design, carrier recombination is reduced by 50%, and efficiency is 1.5-2 percentage points higher than traditional BSF.

Mass production average 23.5% (SunPower 2022), laboratory 24.06% (University of New South Wales), global market share 92% (PV InfoLink 2023), cost <\$0.20/W (BNEF), supporting 80% of overseas ground-mounted power plant installations.

Rear Structure Reconstruction

Passivation layer:

The first step in PERC's rear structure reconstruction is to deposit a layer of aluminum oxide (Al₂O₃) passivation film on the back of the silicon wafer, thickness 10-20nm.

This layer acts like an "anti-leakage film," reducing surface recombination velocity from BSF's 1,000cm/s to <10cm/s.

Besides aluminum oxide, a layer of silicon nitride (SiNx) film (thickness 70-90nm) is added, serving to further passivate + anti-reflect.

SunPower (USA) 2021 report shows, dual-layer structure reduces recombination velocity by another 20% compared to single Al₂O₃ film, corresponding to a 0.3 percentage point efficiency increase.

Local contact:

PERC uses laser grooving + silver paste printing for local contacts.

The laser creates grooves in the passivation film: width 30-50μm, depth 5-8μm (REC Group 2020 process parameters), exposing only 5%-10% of the silicon bulk area, with over 90% still protected by the passivation film.

Compared to BSF's full aluminum contact (100% area contact), local contact reduces metal-silicon direct recombination.

Hanwha Q CELLS (Korea) 2023 test: When contact area proportion is 8%, recombination loss is 40% lower than full contact, efficiency difference 1.2 percentage points.

Optical reflection:

Silicon wafers weakly absorb 900-1200nm light; this light passes through the wafer and exits from the back. PERC uses a silicon nitride film (refractive index 2.0-2.1) to reflect this light back, allowing it to be absorbed again by the silicon.

Solar Frontier (Japan) 2022 experiment: Backside reflectivity increased from BSF's 65% to 88%, light path increased 15%-20%, corresponding quantum efficiency improvement of 8% at 1000nm.

In actual modules, this can increase daily average power generation by 3%-5%.

Data comparison: BSF vs. PERC rear structure differences

Parameter	BSF (Traditional Aluminum Back Surface Field)	PERC (After Rear Structure Reconstruction)	Data Source
Rear Material	Full aluminum layer (thickness 1-2μm)	Al ₂ O ₃ (10-20nm) + SiNx (70-90nm) + local aluminum	Fraunhofer ISE 2023
Surface Recombination Velocity	300-1000cm/s	<10cm/s	PV Tech 2023
Contact Area Proportion	100%	5%-10%	REC Group 2020
Backside Reflectivity (1000nm)	65%	88%	Solar Frontier 2022
Carrier Recombination Loss	25%-30%	10%-15%	NREL 2021 Simulation
Efficiency Contribution	Baseline 21.5%	+ 1.5-2 percentage points (to 23.5%)	IEA PVPS 2023

Process details:

! SunPower (USA): Uses Atomic Layer Deposition (ALD) for Al₂O₃ coating, uniformity ±0.5nm, optimal passivation effect; Laser grooving uses

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picosecond laser, heat-affected zone <2μm, reduces silicon damage.

I **Hanwha Q CELLS (Korea)**: Bifacial PERC uses dual-layer SiNx (refractive index 2.0+2.3), reflectivity increased to 90%, bifaciality from 70% to 75% (2023 mass production data).

I **LG Solar (Korea)**: Local contacts use copper electroplating instead of silver paste, electrode thickness reduced to 15μm, cost decreased \$0.005/W (PV Manufacturing Report 2022).

Actual effect:

Fraunhofer ISE (Germany) 2022 test: A 156mm monocrystalline silicon wafer, BSF structure efficiency 21.2%, after changing to PERC rear structure, efficiency 23.1% – difference 1.9 percentage points, mainly from reduced rear recombination (contribution 1.2 pct) and light reflection gain (0.7 pct).

500MW power plant in Arizona, USA (2021 operation, using SunPower PERC modules) data shows: After rear structure reconstruction, module operating temperature 2-3°C lower than BSF, annual generation 4.2% more, equivalent to 0.045kWh more per watt per year (SEIA monitoring).

Process challenges:

Al₂O₃ film too thin (<10nm) insufficient passivation, too thick (>20nm) prone to cracking.

Fraunhofer ISE found, at a film thickness of 15nm, the recombination velocity is lowest (8cm/s); exceeding 20nm, stress causes film cracking, recombination velocity rebounds to 50cm/s.

Laser groove positioning also matters. REC Group experiment: When groove spacing is 200-250μm, current collection efficiency is highest;

Spacing <150μm causes short circuit, >300μm increases contact resistance, each reducing efficiency by 0.5 percentage points.

Performance Parameters

How much higher is the efficiency really:

On mass production lines, 2023 overseas manufacturers average efficiency **23.5%** (LG Solar Korean production line), highest laboratory efficiency **24.06%**.

Compared to traditional aluminum back-field cells (BSF), PERC is **1.5-2 percentage points** higher – BSF mass production efficiency stuck at 21.5%, equivalent to 10-15W more power per same area module.

TOPCon laboratory efficiency 24.7% (Fraunhofer ISE 2023), mass production 25.8% (Hanwha Q CELLS Korea, 2023):

HJT laboratory **26.81%** (Japan Kaneka, 2022), mass production **25.2%** (Switzerland Meyer Burger).

PERC mass production 23.5%, though behind, has lower cost **\$0.05-0.08/W** (BNEF 2023), suitable for price-sensitive scenarios.

Bifaciality:

PERC bifaciality **75%**, far superior to BSF's **0%**, but lower than TOPCon's **85%+** and HJT's **95%+**.

Germany TÜV Rheinland test: With grass reflectivity 15%, PERC bifacial modules generate **8%** more power than monofacial;

With sandy ground reflectivity 30%, generate **12%** more power.

While HJT on same sandy ground can generate **18%** more, the difference is due to higher bifaciality.

Temperature coefficient:

PERC temperature coefficient **-0.35%/°C**, BSF is **-0.45%/°C**, with the same 10°C temperature rise, PERC efficiency drops 0.1 percentage points less.

500MW power plant in Arizona, USA (SunPower PERC modules, 2021 operation) data: Summer module operating temperature 55°C (ambient 40°C), PERC generates **4.2%** more daily average power than BSF (SEIA monitoring).

Tropical regions (e.g., India), PERC annual generation 3-5% higher than BSF (India MNRE 2023 report).

Light-Induced Degradation (LID):

PERC initial LID **1.5%**, lower than BSF's **2.5%**. But through hydrogen passivation treatment, PERC LID can be reduced to **0.5%**.

Long-term degradation: PERC annual average **0.45%**, similar to TOPCon (0.4%) and HJT (0.38%).

But PERC degrades slightly faster in high-temperature, high-humidity environments (e.g., Southeast Asia), annual average **0.5%**.

Power output:

182mm wafer PERC module mass production power **410-430W**, 210mm wafer **540-560W**.

Compared to BSF same-size modules, 182mm is **35-40W** more, 210mm **50-55W** more.

Fraunhofer Institute Germany test: At irradiance 200W/m² (cloudy day), PERC efficiency maintains **21.8%**, BSF only **19.5%**, HJT **22.5%**.

Cost-effectiveness:

PERC manufacturing cost <**0.18-0.20/W** (BNEF 2023, overseas manufacturers), significantly lower than TOPCon's **0.25-0.28/W** (additional tunnel oxide layer equipment) and HJT's **\$0.30-0.35/W** (low-temperature silver paste expensive).

SunPower US production line data shows: PERC non-silicon cost 0.12/W, TOPCon due to additional cleaning/texturing processes, non-silicon cost **0.17/W**.

US residential solar project (SEIA 2023 case): PERC module system cost **2.8/W**, TOPCon **3.2/W**. Based on annual generation 1,200kWh/kW, PERC earns **\$0.04/W/year** more, recouping cost difference in 6 years.



TOPCon

TOPCon (Tunnel Oxide Passivated Contact) is an N-type cell technology led by overseas photovoltaic companies, mass production efficiency reaches 25.0%-25.8% (NREL 2023), 1.3-2.3 percentage points higher than PERC.

Core is 1-2nm silicon oxide + doped polysilicon layer structure, open-circuit voltage >720mV (ISFH test), bifaciality >85%, temperature coefficient <-0.29%/°C.

2023 global planned capacity exceeds 400GW; European/US companies like REC Group, Q Cells mass production line yield >97%, module power 610W+, electricity cost 3-5% lower than PERC.

Technical Principle

Silicon oxide layer:

I **Thickness determines effect:** ISFH (Institute for Solar Energy Research Hamelin, Germany) test found, with oxide layer thickness 1.2nm, electron tunneling probability approx. 70%, leakage current <1fA/cm²; at 1.8nm, tunneling probability drops to 30%, leakage current rises to 5fA/cm²; exceeding 2nm basically cannot tunnel, becomes ordinary insulating layer.

I **Preparation method affects uniformity:** REC Group uses **Atomic Layer Deposition (ALD)** for oxide layer; thickness deviation controlled within ±0.1nm, more stable than thermal oxidation (deviation ±0.3nm), cell efficiency can differ by 0.3%.

Polysilicon layer:

I **N-type TOPCon** (mainstream): Uses phosphorus (P) doping, polysilicon positively charged, attracts electrons in wafer (N-type silicon has more electrons), forming ohmic contact (resistance <10mΩ·cm², NREL data).

I **P-type TOPCon** (less common): Uses boron (B) doping, negatively charged, attracts holes.

Fraunhofer ISE experiment shows: At phosphorus concentration 1e19 atoms/cm³, contact resistance lowest (8mΩ·cm²), efficiency 25.1%;

Concentration rises to 5e19, resistance slightly increases (12mΩ·cm²), efficiency drops 0.2%.

Bifacial design:

The backside also has 1.5 nm silicon oxide + phosphorus-doped polysilicon, equivalent to adding a "protective shield" to the wafer back.

I **Bifaciality improvement:** Fraunhofer ISE measured bifacial TOPCon bifaciality 88%, PERC only 78%. Backside power gain obvious when ground reflectivity high – e.g., snow environment, backside contribution can reach 25% of total generation.

I **Better low-light performance:** At dawn/dusk weak light, TOPCon short-circuit current 3-5% higher than PERC (NREL simulation), because bifacial structure captures more scattered light.

Carrier behavior:

Electron path in TOPCon is "smooth":

1. Sunlight enters wafer, excites electron-hole pairs

2. Electrons drift to front side, pass through 1.5nm oxide layer (tunneling), enter polysilicon layer;

3. Polysilicon layer conducts electrons to metal electrode (silver grid lines);

4. Holes blocked by oxide layer, remain in wafer drifting to backside electrode

ISFH measured TOPCon surface recombination velocity (SRV) <10 fA/cm². PERC is 50-100 fA/cm².

NREL data shows: TOPCon Voc generally 720-730mV, PERC only 680-695mV, 40mV difference can increase efficiency by 1 percentage point.

Process implementation:

The oxide layer and polysilicon layer must be made in sequence. Overseas companies commonly use two routes:

Step	Method 1 (LPCVD based)	Method 2 (ALD+PECVD)
Oxide Layer Preparation	Thermal oxidation (900°C, 30 min)	ALD (trimethylaluminum + water, low temperature 200°C)
Polysilicon Layer Preparation	LPCVD (phosphine gas, 620°C)	PECVD (silane + phosphine, 400°C)
Advantage	Fast speed (200 wafers/hour)	Thickness uniform (deviation ±0.1nm)
Disadvantage	Oxide layer thickness difficult to control (±0.3nm)	Slower speed (100 wafers/hour)

REC Group chooses Method 2, though slower, yield 98%;

Q Cells uses Method 1, relies on laser edge trimming to compensate thickness deviation, yield 96%.

Comparison with other passivated contacts:

Compared to PERC's Al₂O₃ passivation and HJT's amorphous silicon passivation:

I **PERC:** Al₂O₃ passivation good but contact resistance high (20-30mΩ·cm²), efficiency stuck around 24.5%;

I **HJT:** Amorphous silicon passivation better (SRV <5fA/cm²), but afraid of high temperature (>200°C crystallization), equipment expensive (\$350 million/GW);

I **TOPCon:** Passivation close to HJT (SRV <10fA/cm²), contact resistance low (8-12mΩ·cm²), also compatible with some PERC equipment (e.g., diffusion furnace modified for phosphorus diffusion), overseas companies think "modification cost low, risk small."

Layout and Capacity

Which overseas companies are investing heavily in expanding TOPCon production lines?

Since 2023, photovoltaic companies in Europe, America, and Asia (excluding China) have intensively announced TOPCon capacity plans, aiming to capture the N-type market. Major players include:

I **REC Group** (Norway): Announced 2023 investment \$1.2 billion to expand TOPCon factory in Singapore, phase 1 capacity 6GW, operational 2024; phase 2 add 4GW, full production 2025.

I **Q Cells** (Germany, under Hanwha Group): Korean Chungcheongbuk-do base existing TOPCon cell capacity 3GW, 2024 add 5GW, total capacity 8GW; Georgia, USA factory planning 3GW module capacity, operational 2025.

I **Maxeon** (USA, former SunPower subsidiary): Malaysia Kulim factory 2023 expands TOPCon capacity from 2GW to 5GW, target 7GW 2024; Nuevo León, Mexico factory planning 4GW, trial production 2025.

I **Panasonic** (Japan): Hyogo Himeji factory pilot TOPCon conversion, 2023 completed 1GW PERC to TOPCon, 2024 target efficiency 25.5%, plans 2025 new 3GW dedicated line.

I **First Solar** (USA): Though mainly thin-film, 2023 partnered with Oxford PV to pilot TOPCon+perovskite tandem pilot line in Germany, capacity 0.5GW, 2024 evaluate mass production feasibility.

Where are factories specifically located, how much can they produce per year?

Overseas TOPCon factory locations concentrate in regions with **low electricity cost, convenient logistics, policy-friendly** environment, capacity scale mostly 3-8GW per factory:

Company	Country	Factory Location	Planned Capacity	Operational Time	2023 Actual Output	Module Power
REC Group	Singapore	Tuas Industrial Estate	10GW (two phases)	2024 Phase 1	2GW (trial production)	590-610W
Q Cells	Korea	Chungcheongbuk-do Eumseong-gun	8GW	2024 Phase 2	3GW	615W
Maxeon	Malaysia	Kulim Hi-Tech Park	7GW	2024	4GW	600-625W
Panasonic	Japan	Hyogo Himeji City	4GW (including conversion)	2025 new build	1GW (conversion line)	580-600W

Sharp (Japan) Japan Mie Kameyama City 3GW 2024 0.5GW (trial production) 585W

What's behind the capacity planning numbers?

IRENA (International Renewable Energy Agency) data shows: 2023 European TOPCon module import volume increased 180% year-on-year; the US due to the Inflation Reduction Act (IRA) subsidies, local TOPCon module demand gap reached 5GW.

I **Ground-mounted power plants:** TOPCon bifacial modules in snow environment with reflectivity >40%, backside power gain 25% (Q Cells Norway measurement), suitable for Nordic, Canadian projects.

I **Residential rooftops:** TOPCon module efficiency 25%+, same area installation capacity 15% more than PERC, fits European small roof demand (Maxeon US residential orders proportion 60%).

Have actual mass production efficiency and yield met targets?

2023 overseas companies' TOPCon mass production efficiency generally reached 25.0%-25.8%, yield stable at 95%-98%, some companies exceeded expectations:

I **REC Group Singapore factory:** Q4 2023 mass production efficiency 25.6%, yield 98% (ALD oxide layer technology), module power 610W, bifaciality 87%.

I **Q Cells Korean base:** Cell efficiency 25.4%, module Q, PEAK DUO L-G10+ power 615W, yield 96% (laser SE technology optimized open-circuit voltage).

I **Maxeon Malaysia factory:** Efficiency 25.2%, yield 97%, first-year degradation <1% (low-corrosion metallization process).

What are the different regional capacity emphases?

I **Europe** (25% of global planned capacity): Focus on module packaging; REC Group, Q Cells European factories emphasize high-power modules (600W), meeting local power plant bidding requirements (EU 2023 new regulations require module efficiency >22.5%).

I **North America** (20%): US factories (Q Cells Georgia, Maxeon Mexico) emphasize localized production, receive IRA tax credits (\$0.07 per watt), modules exported to US market tariff-free.

I **Southeast Asia (excluding China)** (35%): Malaysia, Vietnam factories focus on low-cost manufacturing, utilizing local low electricity prices (\$0.05-0.07/kWh), module cost 15% lower than Europe.

I **Japan/Korea** (10%): Japan Panasonic, Sharp focus on technology iteration, testing thinning (130μm silicon wafers) and silver-coated copper paste, target silver consumption below 100mg/W by 2025.

Where does the money for capacity expansion come from?

Company expansion funds mainly from **internal cash flow + government subsidies + bank loans**:

I REC Group Singapore factory received Singapore Economic Development Board (EDB) subsidy \$120 million (10% of total investment);

I Q Cells US factory utilizes IRA subsidies, loan interest as low as 3.5% (supported by US Export-Import Bank);

I Maxeon Malaysia factory funds from parent company TCL Zhonghuan (non-Chinese company) injection \$300 million.

2023 global TOPCon expansion total investment exceeded \$20 billion, of which equipment procurement accounted for 60% (LPCVD, ALD equipment suppliers Applied Materials, Meyer Burger orders scheduled to 2025).

How long does capacity ramp-up take, what are the hurdles?

From factory groundbreaking to full production usually takes 18-24 months, main bottlenecks are **equipment debugging** and **process磨合**:

I REC Group Singapore factory started construction in 2022, trial production Q1 2024, due to ALD equipment delivery delay (German supplier capacity shortage), mass production delayed 3 months;

I Q Cells Korean base phase 2 uses LPCVD equipment, initial oxide layer thickness deviation ±0.3nm, through laser edge trimming technology took 6 months to stabilize to ±0.1nm;

I Yield ramp-up: New factories' first 6 months yield mostly 90%-93%, after 12 months reach above 96% (Maxeon Malaysia factory data).

Currently overseas TOPCon capacity utilization approx. 85% (2023 Q4), mainly affected by European winter construction slowdown, US module inventory backlog, 2024 expected recovery to 90%.



HJT

Laboratory efficiency reached 26.7% (Kaneka, 2017), mass production average 25.1% (REC Group, 2023), temperature coefficient -0.24%/°C, bifaciality 92% (Hevel Solar).

Low-temperature process compatible with 100μm silicon wafers, silicon material saved 30%.

Challenges: Expensive equipment (Meyer Burger full line €150 million/GW), silver paste consumption 180mg/wafer (PERC 120mg), pushing silver-coated copper (Heraeus Ag-Cu paste), electroplated copper to reduce costs.

Layered Design

First layer:

No P-type silicon, because N-type silicon has longer minority carrier lifetime, meaning electrons are less likely to be "tripped" by impurities when moving, less recombination loss.

Silicon wafer thickness now trending thinner; German Q CELLS tested 90μm, breakage rate <0.1% (0.3% at 100μm), but mainstream still 100-130μm.

Wafer resistivity 1-3Ω·cm (Norway REC Group mass production standard); too thick resistance high, too thin prone to break, this range current collection most stable.

Second layer:

First, a layer of **intrinsic amorphous silicon (i-a-Si: H)** is laid on the front and back of the wafer, thickness 5-10nm (Japan Panasonic patent data), equivalent to pasting a "preservative film" on the silicon surface.

Germany Fraunhofer ISE measured: i-a-Si: H interface state density <10¹⁰ cm⁻², 100 times lower than PERC's aluminum back field passivation (~10¹² cm⁻²), so carrier recombination less, open-circuit voltage can increase.

This layer deposited using **PECVD equipment**, temperature below 150°C.

Third layer:

Outside i-a-SiH₄, **doped amorphous silicon** is laid: front side p-type (boron doped, thickness 5-8nm), back side n-type (phosphorus doped, thickness 5-8nm), forming a PN junction with the underlying N-type silicon substrate.

In Japan Panasonic's scheme: p-type layer boron concentration 10¹⁹ cm⁻³, n-type layer phosphorus concentration 5×10¹⁸ cm⁻³, this way built-in electric field strong enough, charge separation efficiency high.

Thickness cannot exceed 10nm; thicker amorphous silicon absorbs more light, and increases series resistance.

Fourth layer:

On top of doped amorphous silicon, a layer of **Transparent Conductive Oxide (TCO)** is laid, mainly indium tin oxide (ITO), also tested with iodine tungsten oxide (IWO).

USA Applied Materials equipment deposits ITO, thickness 80-100nm, transmittance >90% (wavelength 300-1100nm), resistivity <3×10⁻⁴ Ω·cm.

Indium is a rare metal; currently testing indium-free ZnO: Al (USA First Solar scheme), transmittance 92%, resistivity slightly higher (5×10⁻⁴ Ω·cm), but cost low.

Fifth layer:

Topmost is metal grid line electrode, using silver paste printing (low-temperature silver paste, melting point <200°C), line width 30-40μm (Meyer Burger screen printing machine parameters), spacing 1.2mm.

Silver paste consumption currently 180mg/wafer (PERC only 120mg), because HJT is bifacial generation, both sides need electrodes.

Switzerland Meyer Burger testing **electroplated copper electrode**, silver consumption reduced to <20mg/wafer, contact resistance even lower (2mΩ·cm² vs silver paste's 4mΩ·cm²), but equipment expensive, one line extra cost €30 million.

The "coordination account" in the layered design:

E.g., if i-a-Si: H is too thick, light absorbed by it, wafer absorbs less light; too thin, passivation insufficient, recombination increases again.

Norway REC Group mass production data: When i-a-Si: H thickness 8nm, p/n amorphous silicon each 6nm, ITO thickness 90nm, efficiency highest (25.1%).

USA NREL simulation: Total layer thickness controlled within 100-120nm (excluding silicon substrate), light loss minimized, carrier transport fastest.

Performance

How high is the efficiency really:

I **Laboratory efficiency**: Japan Kaneka achieved 26.7% in 2017 (area 180cm²), later European institutions followed; Fraunhofer ISE 2022 used "microcrystalline silicon + amorphous silicon" layers, efficiency reached 26.8%; 2023 USA NREL using HJT as bottom cell, perovskite tandem efficiency exceeded 32%, but pure HJT still around 26.7%.

I **Mass production efficiency**: Norway REC Group 2023 mass production line average 25.1%, highest batch 25.3% (module area 2.58m²); Switzerland Meyer Burger G10 line mass production efficiency 25.0%, yield above 95%; USA Hevel Solar HJT modules, third-party certified efficiency 24.9%.

I Comparison: Same period PERC mass production efficiency 23.5%-24.0% (JinkoSolar 2023 annual report), TOPCon 24.5%-25.0% (LG Energy Solution data), HJT indeed 0.5-1 percentage point higher.

Does power generation drop fast when it's hot:

I **Temperature coefficient**: HJT is -0.24%/°C, PERC is -0.35%/°C, TOPCon is -0.30%/°C (NREL 2023 report). Meaning: For every 1°C increase in ambient temperature, HJT power drops 0.24%, PERC drops 0.35%.

I **Actual measurement comparison**: Phoenix, Arizona, USA summer (average temperature 42°C, module surface 45°C), HJT module power 12% higher than PERC, 8% higher than TOPCon (REC Group 500MW project monitoring data); Munich, Germany summer (35°C), HJT generates 9% more power than PERC.

I Conversely, when cold in winter, HJT is also stable: -0.24%/°C coefficient means less power loss at low temperatures; Berlin, Germany winter (5°C), HJT module generation 6% higher than TOPCon.

How much can the backside generate:

I **Bifaciality**: REC Group mass production module bifaciality 92% (TÜV Süd certification), Meyer Burger 91%, Hevel Solar 93%; PERC generally 78%-82% (JinkoSolar), TOPCon 84%-87% (LG).

I **Backside gain**: In ground-mounted power plant scenarios, HJT backside generation gain 10%-30% (depending on ground reflectivity). E.g., 500MW project in California, USA, ground reflective sand (reflectivity 30%), HJT backside generates 28% more; REC Group project in Sweden (snow reflectivity 50%), backside gain 31%.

Does efficiency drop after long-term use:

I **First-year degradation**: HJT <1% (REC Group measured 0.8%), PERC 2% (JinkoSolar), TOPCon 1.5% (LG). Reason: HJT has no LID (Light-Induced Degradation) and LeTID (Light and elevated Temperature Induced Degradation); PERC/TOPCon boron-oxygen recombination causes initial degradation.

I **Long-term degradation**: Over 25-year lifespan, HJT annual average degradation 0.4%-0.5% (Fraunhofer ISE model), PERC 0.5%-0.6%, TOPCon 0.45%-0.55%. Calculated: After 25 years, HJT module efficiency remains 88%, PERC 83%, TOPCon 86%.

How is generation on cloudy days or dawn/dusk:

I **Low-light efficiency**: At irradiance 200W/m² (cloudy day or dawn/dusk), HJT efficiency maintains 92% of initial value, PERC 85%, TOPCon 88% (NREL low-light test).

I **Daily average generation hours**: Distributed project in Hamburg, Germany (2022 installed HJT), daily average generation 4.2 hours (including dawn/dusk low light); PERC modules installed same period only 3.8 hours, annually generates 10% more.

Does power density have an advantage:

I **Module power**: REC Group HJT module (area 2.58m²) power 640W, power density 248W/m²; same area PERC module (JinkoSolar) 600W, density 233W/m²; TOPCon (LGE) 620W, density 240W/m².

I **Rooftop application**: Residential roof in Berlin, Germany (usable area 50m²), installing HJT can install 12.4kW, PERC only 11.65kW, annually generates approx. 700kWh more (based on 1,000 hours sunshine).

Cost Reduction

How to solve expensive equipment:

I Convert second-hand lines: Italy's Elettronova converted old PERC lines (original German Centrotherm) to HJT-compatible lines, replaced high-temperature diffusion furnaces, added PECVD (USA Applied Materials equipment) and low-temperature silver paste printing machines, investment reduced to €105 million/GW, down 30%. But initial yield after conversion was only 92% (new lines 95%), took half a year debugging to catch up.

I Equipment localization attempts: Korea's Hanwha Solution bought Chinese Jiejia Weichuang's PECVD equipment (non-core modules domestic-made), combined with own screen printing machines, full line investment €120 million/GW, 20% cheaper than Meyer Burger.

Silver paste too expensive, how to save overseas:

HJT silver paste consumption 180mg/wafer (PERC 120mg), main reasons: double-sided electrodes + low-temperature silver paste poor conductivity.

Overseas companies mainly focus on two directions:

I Silver-coated copper paste: Germany's Heraeus 2023 launched Ag-Cu paste (silver content 30%, copper 70%), cost 40% lower than pure silver paste, consumption reduced to 150mg/wafer (copper replaces part silver). Switzerland's Meyer Burger trial use, contact resistance increased from 4mΩ·cm² to 4.5mΩ·cm² (acceptable range), module efficiency only dropped 0.1%. USA DuPont also testing silver content 25% version, target consumption 140mg/wafer.

I Electroplated copper electrodes: Switzerland Meyer Burger G10 line added electroplating equipment, silver consumption directly driven to <20mg/wafer (copper replaces most silver), contact resistance reduced to 2mΩ·cm² (better than silver paste). But electroplating line expensive, one line extra cost €30 million, and wastewater treatment cost high (contains copper ions). Norway REC Group plans to use in new lines 2025, expects silver cost per watt reduced from 0.03 to 0.005.

Indium too expensive, change material possible:

TCO layer uses indium tin oxide (ITO), indium price \$300/kg (2023 fluctuation), accounts for 40% TCO cost. Overseas companies testing two substitutes:

I Indium-free TCO: USA First Solar uses ZnO: Al (aluminum-doped zinc oxide) instead of ITO, transmittance 92% (ITO 90%), resistivity 5×10⁻⁴ Ω·cm (ITO 3×10⁻⁴ Ω·cm), efficiency maintained 24.8% (slightly dropped 0.3%). Disadvantage: conductivity slightly worse, needs thickening to 120nm (ITO 80nm).

I Low-indium formulation: Japan Tosoh Chemical developed ITO with 10% tin (ITO-Sn), indium usage reduced 20%, transmittance and resistivity unchanged. Germany Schott used it for TCO glass, cost down 15%, already supplied to REC Group for trial.

Thinning really saves silicon material?

Silicon wafer accounts for 30% HJT cost, thinning is key. Germany Q CELLS 2023 tested 90μm wafers (mainstream 100-130μm), silicon material saved 30% (by area), but many problems:

I Breakage rate: 90μm breakage rate <0.1% (0.3% at 100μm), relying on Germany Exatron's laser cutting equipment (kerf width 20μm less);

I Warping: Thin wafers prone to warp, Q CELLS uses "vacuum chuck transfer system" (Italy Baccini equipment), warping controlled within 0.5mm (industry standard 1mm);

I Efficiency loss: 90μm wafer minority carrier lifetime slightly reduced (from 2ms to 1.8ms), efficiency drops 0.2%, but material cost saving more valuable than efficiency loss (silicon material price 8/kg, 90μm saves 0.01/W).

Low yield dragging behind, how overseas factories adjust:

HJT yield lower than PERC (PERC 98%, HJT 95%), main reason amorphous silicon film thickness non-uniform. Overseas factories use these methods to improve yield:

I PECVD process optimization: USA Applied Materials upgraded equipment, uses "multi-chamber independent temperature control", film thickness uniformity improved from ±3nm to ±1.5nm, REC Group new line yield increased to 96%;

I Online monitoring: Germany Meyer Burger adds laser thickness gauge after TCO deposition (Israel Orbotech equipment), real-time target power adjustment, film thickness deviation exceeding 2nm automatically stops;

I Personnel training: Norway REC Group sends engineers to Japan Panasonic factory for 3 months, master amorphous silicon deposition parameter fine-tuning (e.g., gas flow ±0.1sccm adjustment), yield increased from 95% to 97%.

Supply chain immature, how overseas fill gap:

HJT supply chain accessories few, delivery slow. Overseas companies respond this way:

I Blind local suppliers: USA Hovel Solar signed exclusive agreement with Guardian Glass, TCO glass delivery cycle shortened from 8 weeks to 4 weeks (PERC glass 2 weeks);

I Stockpile in advance: Switzerland Meyer Burger stocks indium targets for 3 months inventory (indium price fluctuates heavily), avoid supply disruption;

I Develop secondary suppliers: Germany Q CELLS introduces Japan Asahi Glass as TCO glass alternative, price 5% lower than Guardian, but transmittance slightly worse (89% vs 90%).

